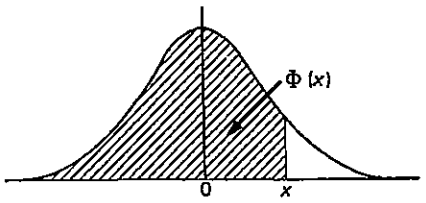
# TABLE 4. THE NORMAL DISTRIBUTION FUNCTION

The function tabulated is $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}t^2}\, dt$. $\Phi(x)$ is the probability that a random variable, normally distributed with zero mean and unit variance, will be less than or equal to $x$. When $x < 0$ use $\Phi(x) = 1 - \Phi(-x)$, as the normal distribution with zero mean and unit variance is symmetric about zero.




| $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0·00 | 0·5000 | 0·40 | 0·6554 | 0·80 | 0·7881 | 1·20 | 0·8849 | 1·60 | 0·9452 | 2·00 | 0·97725 |
| ·01 | ·5040 | ·41 | ·6591 | ·81 | ·7910 | ·21 | ·8869 | ·61 | ·9463 | ·01 | ·97778 |
| ·02 | ·5080 | ·42 | ·6628 | ·82 | ·7939 | ·22 | ·8888 | ·62 | ·9474 | ·02 | ·97831 |
| ·03 | ·5120 | ·43 | ·6664 | ·83 | ·7967 | ·23 | ·8907 | ·63 | ·9484 | ·03 | ·97882 |
| ·04 | ·5160 | ·44 | ·6700 | ·84 | ·7995 | ·24 | ·8925 | ·64 | ·9495 | ·04 | ·97932 |
| 0·05 | 0·5199 | 0·45 | 0·6736 | 0·85 | 0·8023 | 1·25 | 0·8944 | 1·65 | 0·9505 | 2·05 | 0·97982 |
| ·06 | ·5239 | ·46 | ·6772 | ·86 | ·8051 | ·26 | ·8962 | ·66 | ·9515 | ·06 | ·98030 |
| ·07 | ·5279 | ·47 | ·6808 | ·87 | ·8078 | ·27 | ·8980 | ·67 | ·9525 | ·07 | ·98077 |
| ·08 | ·5319 | ·48 | ·6844 | ·88 | ·8106 | ·28 | ·8997 | ·68 | ·9535 | ·08 | ·98124 |
| ·09 | ·5359 | ·49 | ·6879 | ·89 | ·8133 | ·29 | ·9015 | ·69 | ·9545 | ·09 | ·98169 |
| 0·10 | 0·5398 | 0·50 | 0·6915 | 0·90 | 0·8159 | 1·30 | 0·9032 | 1·70 | 0·9554 | 2·10 | 0·98214 |
| ·11 | ·5438 | ·51 | ·6950 | ·91 | ·8186 | ·31 | ·9049 | ·71 | ·9564 | ·11 | ·98257 |
| ·12 | ·5478 | ·52 | ·6985 | ·92 | ·8212 | ·32 | ·9066 | ·72 | ·9573 | ·12 | ·98300 |
| ·13 | ·5517 | ·53 | ·7019 | ·93 | ·8238 | ·33 | ·9082 | ·73 | ·9582 | ·13 | ·98341 |
| ·14 | ·5557 | ·54 | ·7054 | ·94 | ·8264 | ·34 | ·9099 | ·74 | ·9591 | ·14 | ·98382 |
| 0·15 | 0·5596 | 0·55 | 0·7088 | 0·95 | 0·8289 | 1·35 | 0·9115 | 1·75 | 0·9599 | 2·15 | 0·98422 |
| ·16 | ·5636 | ·56 | ·7123 | ·96 | ·8315 | ·36 | ·9131 | ·76 | ·9608 | ·16 | ·98461 |
| ·17 | ·5675 | ·57 | ·7157 | ·97 | ·8340 | ·37 | ·9147 | ·77 | ·9616 | ·17 | ·98500 |
| ·18 | ·5714 | ·58 | ·7190 | ·98 | ·8365 | ·38 | ·9162 | ·78 | ·9625 | ·18 | ·98537 |
| ·19 | ·5753 | ·59 | ·7224 | ·99 | ·8389 | ·39 | ·9177 | ·79 | ·9633 | ·19 | ·98574 |
| 0·20 | 0·5793 | 0·60 | 0·7257 | 1·00 | 0·8413 | 1·40 | 0·9192 | 1·80 | 0·9641 | 2·20 | 0·98610 |
| ·21 | ·5832 | ·61 | ·7291 | ·01 | ·8438 | ·41 | ·9207 | ·81 | ·9649 | ·21 | ·98645 |
| ·22 | ·5871 | ·62 | ·7324 | ·02 | ·8461 | ·42 | ·9222 | ·82 | ·9656 | ·22 | ·98679 |
| ·23 | ·5910 | ·63 | ·7357 | ·03 | ·8485 | ·43 | ·9236 | ·83 | ·9664 | ·23 | ·98713 |
| ·24 | ·5948 | ·64 | ·7389 | ·04 | ·8508 | ·44 | ·9251 | ·84 | ·9671 | ·24 | ·98745 |
| 0·25 | 0·5987 | 0·65 | 0·7422 | 1·05 | 0·8531 | 1·45 | 0·9265 | 1·85 | 0·9678 | 2·25 | 0·98778 |
| ·26 | ·6026 | ·66 | ·7454 | ·06 | ·8554 | ·46 | ·9279 | ·86 | ·9686 | ·26 | ·98809 |
| ·27 | ·6064 | ·67 | ·7486 | ·07 | ·8577 | ·47 | ·9292 | ·87 | ·9693 | ·27 | ·98840 |
| ·28 | ·6103 | ·68 | ·7517 | ·08 | ·8599 | ·48 | ·9306 | ·88 | ·9699 | ·28 | ·98870 |
| ·29 | ·6141 | ·69 | ·7549 | ·09 | ·8621 | ·49 | ·9319 | ·89 | ·9706 | ·29 | ·98899 |
| 0·30 | 0·6179 | 0·70 | 0·7580 | 1·10 | 0·8643 | 1·50 | 0·9332 | 1·90 | 0·9713 | 2·30 | 0·98928 |
| ·31 | ·6217 | ·71 | ·7611 | ·11 | ·8665 | ·51 | ·9345 | ·91 | ·9719 | ·31 | ·98956 |
| ·32 | ·6255 | ·72 | ·7642 | ·12 | ·8686 | ·52 | ·9357 | ·92 | ·9726 | ·32 | ·98983 |
| ·33 | ·6293 | ·73 | ·7673 | ·13 | ·8708 | ·53 | ·9370 | ·93 | ·9732 | ·33 | ·99010 |
| ·34 | ·6331 | ·74 | ·7704 | ·14 | ·8729 | ·54 | ·9382 | ·94 | ·9738 | ·34 | ·99036 |
| 0·35 | 0·6368 | 0·75 | 0·7734 | 1·15 | 0·8749 | 1·55 | 0·9394 | 1·95 | 0·9744 | 2·35 | 0·99061 |
| ·36 | ·6406 | ·76 | ·7764 | ·16 | ·8770 | ·56 | ·9406 | ·96 | ·9750 | ·36 | ·99086 |
| ·37 | ·6443 | ·77 | ·7794 | ·17 | ·8790 | ·57 | ·9418 | ·97 | ·9756 | ·37 | ·99111 |
| ·38 | ·6480 | ·78 | ·7823 | ·18 | ·8810 | ·58 | ·9429 | ·98 | ·9761 | ·38 | ·99134 |
| ·39 | ·6517 | ·79 | ·7852 | ·19 | ·8830 | ·59 | ·9441 | ·99 | ·9767 | ·39 | ·99158 |
| 0·40 | 0·6554 | 0·80 | 0·7881 | 1·20 | 0·8849 | 1·60 | 0·9452 | 2·00 | 0·9772 | 2·40 | 0·99180 |

# TABLE 4. THE NORMAL DISTRIBUTION FUNCTION

| $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ | $x$ | $\Phi(x)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2·40 | 0·99180 | 2·55 | 0·99461 | 2·70 | 0·99653 | 2·85 | 0·99781 | 3·00 | 0·99865 | 3·15 | 0·99918 |
| ·41 | ·99202 | ·56 | ·99477 | ·71 | ·99664 | ·86 | ·99788 | ·01 | ·99869 | ·16 | ·99921 |
| ·42 | ·99224 | ·57 | ·99492 | ·72 | ·99674 | ·87 | ·99795 | ·02 | ·99874 | ·17 | ·99924 |
| ·43 | ·99245 | ·58 | ·99506 | ·73 | ·99683 | ·88 | ·99801 | ·03 | ·99878 | ·18 | ·99926 |
| ·44 | ·99266 | ·59 | ·99520 | ·74 | ·99693 | ·89 | ·99807 | ·04 | ·99882 | ·19 | ·99929 |
| 2·45 | 0·99286 | 2·60 | 0·99534 | 2·75 | 0·99702 | 2·90 | 0·99813 | 3·05 | 0·99886 | 3·20 | 0·99931 |
| ·46 | ·99305 | ·61 | ·99547 | ·76 | ·99711 | ·91 | ·99819 | ·06 | ·99889 | ·21 | ·99934 |
| ·47 | ·99324 | ·62 | ·99560 | ·77 | ·99720 | ·92 | ·99825 | ·07 | ·99893 | ·22 | ·99936 |
| ·48 | ·99343 | ·63 | ·99573 | ·78 | ·99728 | ·93 | ·99831 | ·08 | ·99896 | ·23 | ·99938 |
| ·49 | ·99361 | ·64 | ·99585 | ·79 | ·99736 | ·94 | ·99836 | ·09 | ·99900 | ·24 | ·99940 |
| 2·50 | 0·99379 | 2·65 | 0·99598 | 2·80 | 0·99744 | 2·95 | 0·99841 | 3·10 | 0·99903 | 3·25 | 0·99942 |
| ·51 | ·99396 | ·66 | ·99609 | ·81 | ·99752 | ·96 | ·99846 | ·11 | ·99906 | ·26 | ·99944 |
| ·52 | ·99413 | ·67 | ·99621 | ·82 | ·99760 | ·97 | ·99851 | ·12 | ·99910 | ·27 | ·99946 |
| ·53 | ·99430 | ·68 | ·99632 | ·83 | ·99767 | ·98 | ·99856 | ·13 | ·99913 | ·28 | ·99948 |
| ·54 | ·99446 | ·69 | ·99643 | ·84 | ·99774 | ·99 | ·99861 | ·14 | ·99916 | ·29 | ·99950 |
| 2·55 | 0·99461 | 2·70 | 0·99653 | 2·85 | 0·99781 | 3·00 | 0·99865 | 3·15 | 0·99918 | 3·30 | 0·99952 |

The critical table below gives on the left the range of values of $x$ for which $\Phi(x)$ takes the value on the right, correct to the last figure given; in critical cases, take the upper of the two values of $\Phi(x)$ indicated.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3·075 | 0·9990 | 3·263 | 0·9994 | 3·731 | 0·99990 | 3·916 | 0·99995 |
| 3·105 | 0·9991 | 3·320 | 0·9995 | 3·759 | 0·99991 | 3·976 | 0·99996 |
| 3·138 | 0·9992 | 3·389 | 0·9996 | 3·791 | 0·99992 | 4·055 | 0·99997 |
| 3·174 | 0·9993 | 3·480 | 0·9997 | 3·826 | 0·99993 | 4·173 | 0·99998 |
| 3·215 | 0·9994 | 3·615 | 0·9998 | 3·867 | 0·99994 | 4·417 | 0·99999 |
| | | | 0·9999 | | 0·99995 | | 1·00000 |

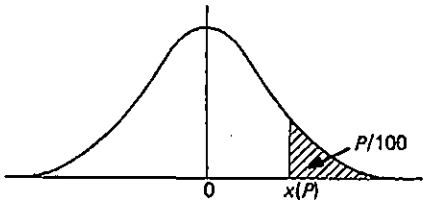
When $x > 3{\cdot}3$ the formula $1-\Phi(x) \doteqdot \frac{e^{-\frac{1}{2}x^2}}{x\sqrt{2\pi}}\left[1-\frac{1}{x^2}+\frac{3}{x^4}-\frac{15}{x^6}+\frac{105}{x^8}\right]$ is very accurate, with relative error less than $945/x^{10}$.

# TABLE 5. PERCENTAGE POINTS OF THE NORMAL DISTRIBUTION

This table gives percentage points $x(P)$ defined by the equation

$$\frac{P}{100} = \frac{1}{\sqrt{2\pi}}\int_{x(P)}^{\infty} e^{-\frac{1}{2}t^2}\,dt.$$

If $X$ is a variable, normally distributed with zero mean and unit variance, $P/100$ is the probability that $X \geqslant x(P)$. The lower $P$ per cent points are given by symmetry as $-x(P)$, and the probability that $|X| \geqslant x(P)$ is $2P/100$.




| $P$ | $x(P)$ | $P$ | $x(P)$ | $P$ | $x(P)$ | $P$ | $x(P)$ | $P$ | $x(P)$ | $P$ | $x(P)$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 50 | 0·0000 | 5·0 | 1·6449 | 3·0 | 1·8808 | 2·0 | 2·0537 | 1·0 | 2·3263 | 0·10 | 3·0902 |
| 45 | 0·1257 | 4·8 | 1·6646 | 2·9 | 1·8957 | 1·9 | 2·0749 | 0·9 | 2·3656 | 0·09 | 3·1214 |
| 40 | 0·2533 | 4·6 | 1·6849 | 2·8 | 1·9110 | 1·8 | 2·0969 | 0·8 | 2·4089 | 0·08 | 3·1559 |
| 35 | 0·3853 | 4·4 | 1·7060 | 2·7 | 1·9268 | 1·7 | 2·1201 | 0·7 | 2·4573 | 0·07 | 3·1947 |
| 30 | 0·5244 | 4·2 | 1·7279 | 2·6 | 1·9431 | 1·6 | 2·1444 | 0·6 | 2·5121 | 0·06 | 3·2389 |
| 25 | 0·6745 | 4·0 | 1·7507 | 2·5 | 1·9600 | 1·5 | 2·1701 | 0·5 | 2·5758 | 0·05 | 3·2905 |
| 20 | 0·8416 | 3·8 | 1·7744 | 2·4 | 1·9774 | 1·4 | 2·1973 | 0·4 | 2·6521 | 0·01 | 3·7190 |
| 15 | 1·0364 | 3·6 | 1·7991 | 2·3 | 1·9954 | 1·3 | 2·2262 | 0·3 | 2·7478 | 0·005 | 3·8906 |
| 10 | 1·2816 | 3·4 | 1·8250 | 2·2 | 2·0141 | 1·2 | 2·2571 | 0·2 | 2·8782 | 0·001 | 4·2649 |
| 5 | 1·6449 | 3·2 | 1·8522 | 2·1 | 2·0335 | 1·1 | 2·2904 | 0·1 | 3·0902 | 0·0005 | 4·4172 |